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Design and Implementation FTTH GPON Network Design Based on Attenuation Measurement and Optisystem Modeling

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ABSTRACT

This study aims to design and implement a Fiber to the Home (FTTH) network based on Gigabit Passive Optical Network (GPON) technology, using attenuation measurement to detect physical disturbances in optical fiber through Optisystem modeling. The research was conducted in a laboratory setting, involving stages of network topology design, system simulation using Optisystem software, implementation of an FTTH trainer module, and attenuation measurement using an Optical Power Meter (OPM). Testing was performed under various cable conditions, including normal, bending, damaged core, and different dropcore cable lengths to evaluate their impact on attenuation. The results indicate that attenuation under normal conditions ranged from 14.94 to 16.23 dB, while physical disturbances such as bending and core damage caused a significant increase in attenuation values. Additionally, increasing cable length contributed to attenuation growth of approximately 0.9 dB for every 5 meters. The simulation results obtained from Optisystem showed strong agreement with link power budget calculations and actual measurements, with a deviation of around 0.1 dB. Severe cable damage, such as a broken core, produced attenuation values exceeding 50 dB, which can serve as a clear indicator of physical disturbances. Overall, this study demonstrates that attenuation measurement combined with Optisystem modeling is effective for detecting physical disturbances in FTTH GPON networks. The developed trainer module also provides a practical learning tool for understanding attenuation characteristics and optical network performance in a laboratory environment.

INTRODUCTION

In the current era of rapid advancements in science and technology, the use of copper cables is no longer considered effective for supporting high-capacity and high-speed bandwidth demands. As a solution to these limitations, optical fiber has emerged as a widely adopted transmission medium to replace copper cables (Yustini et al., 2021). Fiber-to-the-Home (FTTH) technology has become a crucial component of modern communication infrastructure, offering high data rates, large bandwidth capacity, and improved stability. Therefore, a comprehensive understanding of its operating principles, installation processes, and measurement techniques is essential for students and practitioners in the field of telecommunications.

However, the use of optical fiber as a transmission medium is not without challenges. One of the primary issues is attenuation along the fiber, which can lead to information loss and a reduction in signal power levels. Excessive attenuation in an optical fiber link can significantly disrupt the required power stability (Farhan et al., 2024). In addition, optical fiber transmission is often affected by various technical issues, such as fiber core breakage, contaminated adapters, and dirty connectors. These factors may introduce additional losses, ultimately degrading the quality of the received signal.

As reported in (Ramadan et al., 2024), the performance of an FTTH network can be evaluated by comparing field measurement results with simulation outputs generated using OptiSystem software. The evaluation focuses on three key parameters: rise time budget, Bit Error Rate (BER), and Signal-to-Noise Ratio (SNR). The simulation results indicate that all three parameters meet the performance standards of GPON technology, confirming that the FTTH design is feasible for implementation to support Triple Play services by PT Telekomunikasi Indonesia Tbk. The study also includes an analysis of the FTTH network design in Surau Gadang Village based on standards established by PT ICON+.

The performance of a network is strongly influenced by the level of attenuation, with an acceptable range typically between 15 dB and 28 dB. To determine whether an FTTH network is suitable for operation according to established standards, quality testing is conducted, focusing on Quality of Service (QoS) analysis (Mukiman & Habib, 2023)

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Currently, the Fiber Optic Communication Systems Laboratory provides several trainer modules for attenuation measurement analysis, including modules for measuring cable attenuation with or without splicing, as well as attenuation caused by passive splitters. However, there is no available equipment or facility for comprehensive FTTH network measurement that covers the entire system architecture—from the service provider side (Optical Line Terminal/OLT) to the user side (Optical Network Unit/ONU). Furthermore, adequate FTTH network activation facilities to support indoor practicum activities and Quality of Service (QoS) measurements are not yet available. This limitation presents a significant challenge in the learning process, particularly for courses that require simulation or direct implementation of FTTH network measurements.

Therefore, a project is required to develop an FTTH network within the SKSO laboratory as a practicum facility. The design of the FTTH network system encompasses the entire architecture, from the service provider side (Optical Line Terminal/OLT) to the user side (Optical Network Unit/ONU), along with its performance measurement capabilities. This study aims to provide a simulation environment that closely resembles real-world conditions, enabling students to perform direct measurement and analysis of FTTH networks, thereby significantly enhancing the quality of practical learning.

LITERATURE REVIEW

Fiber To The Home (FTTH) is a network deployment model in which optical fiber is extended from the central office directly to the subscriber's premises. This technology is widely recognized for its capability to deliver high bandwidth with relatively low transmission losses . One of the most widely adopted standards in FTTH implementation is Gigabit Passive Optical Network (GPON), standardized by ITU-T G.984, which enables the delivery of Triple Play services (voice, video, and data) over a single optical fiber infrastructure. Several studies have been conducted to analyze and improve the performance of FTTH networks, particularly those based on GPON technology designed an FTTH network using OptiSystem software to simulate network performance, including link power budget, Bit Error Rate (BER), and Q-Factor. The results indicated that the designed network met the minimum performance standards and was considered feasible for implementation. Similarly, Amalia Rizqi Utami et al. (Utami et al., 2022) utilized OptiSystem simulation combined with link power budget analysis to evaluate FTTH network performance. Both studies emphasize the effectiveness of simulation tools in analyzing system performance; however, their focus remains limited to performance evaluation without addressing the detection of physical disturbances in the network.

In addition to simulation-based approaches, several studies have focused on empirical and analytical analysis of attenuation in FTTH networks. (Mukiman & Habib, 2023) investigated attenuation reporting systems in GPON-based FTTH networks using the link power budget method. Their findings highlight that attenuation parameters can serve as indicators of network disturbances, with contributing factors such as splice losses, dirty connectors, and fiber bending. Similarly, (Sidiq et al., 2025) analyzed attenuation in an FTTH GPON network in an urban environment, identifying issues such as bending losses, poor splicing, and network branching as major contributors to increased attenuation.

Further research by (Reynaldi et al., 2024) examined attenuation in the distribution segment of FTTH networks (ODC to ODP) and emphasized the importance of monitoring attenuation values to maintain network quality. However, the study primarily focused on identifying attenuation thresholds and their causes, without incorporating simulation-based modeling or systematic disturbance detection methods. Likewise, Sondang Sibuea et al. (2022) (Sidiq et al., 2025)utilized OTDR and link power budget calculations to evaluate network attenuation, confirming that the network met standard requirements. Nevertheless, the study was limited to measurement-based analysis without integrating simulation tools for comprehensive system modeling.

From an implementation and educational perspective, (Fauzi & Subagja, 2021) discussed the configuration of FTTH networks as a broadband solution and emphasized its advantages over conventional copper-based systems, such as higher bandwidth capacity, lower attenuation, and resistance to electromagnetic interference. Meanwhile, (Sidiq et al., 2025) highlighted the need for practical learning media in optical fiber communication laboratories, noting that existing training modules are still limited and do not fully support hands-on understanding of attenuation phenomena and network performance. Based on the reviewed literature, it can be observed that previous studies generally focus on three main aspects (Sidiq et al., 2025)network performance simulation using software such as OptiSystem, (Mukiman & Habib, 2023)empirical and analytical attenuation analysis using link power budget and OTDR measurements, and implementation and educational applications of FTTH networks. However, there remains a research gap in the integration of simulation-based modeling with empirical attenuation measurements for the purpose of systematically detecting physical disturbances in FTTH GPON networks.

Therefore, this study aims to address this gap by combining attenuation measurement methods with OptiSystem-based modeling to detect physical disturbances in FTTH GPON networks. This approach is expected to provide a more comprehensive analysis by integrating theoretical calculations, simulation results, and empirical measurements, thereby improving the accuracy of network performance evaluation and disturbance detection.

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METHOD

This study employs an experimental and simulation-based approach to design and implement a Fiber-to-the-Home (FTTH) network in a laboratory environment. The research focuses on evaluating network performance based on attenuation measurements and Quality of Service (QoS) parameters. In addition, simulations are conducted using OptiSystem software to validate the experimental results.

Research Flow

The research stages are carried out as follows:

- Literature Study
  Reviewing references related to FTTH networks, GPON technology, attenuation measurements, and QoS parameters.
- System Design
  Designing the FTTH network architecture, which includes the Optical Line Terminal (OLT), Optical Distribution Network (ODN), and Optical Network Unit (ONU).
- Simulation Using OptiSystem Modeling
  Performing network modeling to analyze performance parameters such as Bit Error Rate (BER), Signal-to-Noise Ratio (SNR), and rise time budget.
- Network Implementation
  Constructing the FTTH network in the laboratory using available devices and trainer modules.
- Measurement and Data Collection
  Measuring attenuation values using an Optical Power Meter and evaluating QoS parameters.
- Data Analysis
  Analyzing the collected data.
- Comparison
  Comparing measurement results with simulation outcomes and applicable standards.
- Conclusion and Evaluation
  Drawing conclusions regarding the feasibility and performance of the network.

System Design

- Modeling Design Using OptiSystem
  Designing and simulating the FTTH network model using OptiSystem software.

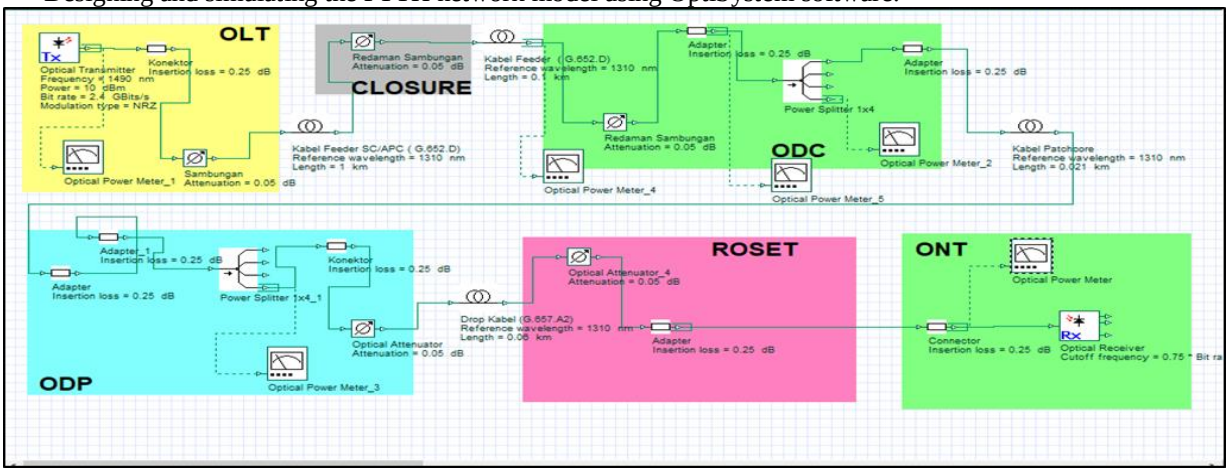


Figure 1. Design of the trainer module in the Optisystem application

- Practicum Trainer Module Design
  Designing a trainer module to support practical laboratory activities.

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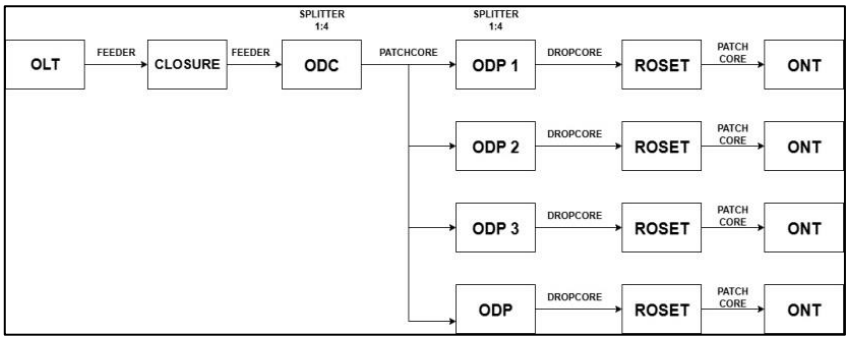


Figure 2. Practicum Trainer module design

- Tools and Materials
  The equipment used in this study includes:
  a. Optical Line Terminal (OLT)
  b. Optical Network Unit (ONU)
  c. Fiber optic cables (G.652D / G.657A)
  d. Passive splitter
  e. Optical Power Meter (OPM)
  f. Optical Light Source (OLS)
  g. OptiSystem software

- Data Collection Method
  Data are obtained through:
  a. Direct measurement using an Optical Power Meter for attenuation
  b. Simulation data generated from OptiSystem software
  c. Observation of network performance during operation.

RESULT

OptiSystem Modeling Results

Before developing the trainer module, a simulation was first conducted to estimate the attenuation values that would be obtained during actual measurements using OptiSystem application modeling.

After completing the simulation, calculations were performed using analytical formulas to validate the consistency between the simulation results and the actual measurements obtained from the trainer module.

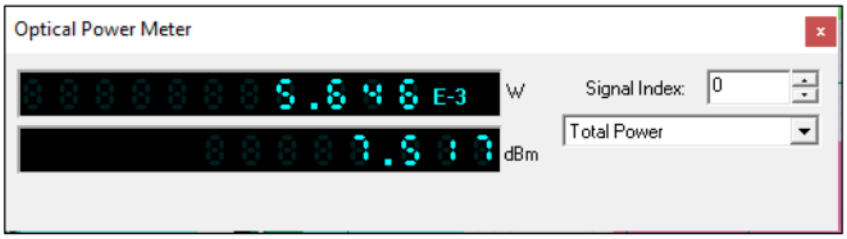


Figure 3. Results Power Transmit OLT measurements on the Optisystem

Measurements using an Optical Power Meter (OPM) at the output of the Optical Line Terminal (OLT) show a transmitted power (Pt) value of 7.517 dBm, as illustrated in Figure 1. The optical signal is then transmitted through the feeder cable and passes through several connection points, such as connectors, splices, and splitters, each of which contributes a certain amount of attenuation to the signal.

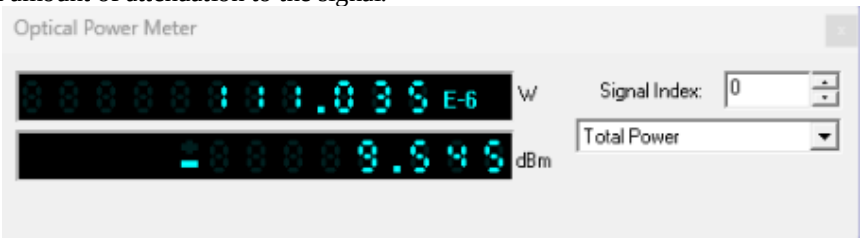


Figure 4. Results Received Power measurements on the Optisystem

At the optical rosette output point, the OPM measurement results show that the received power (Pr) at the ONT

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side, at a wavelength of 1310 nm, is -9.545 dBm after passing through the entire distribution path, as illustrated in Figure 2.

Link budget calculations were carried out using Equations (1), (2), and (3) to ensure the compatibility between simulation results and theoretical calculations. Table 1 presents the attenuation values associated with each device used in the system.

Table 1. Attenuation values on the device

Parameter		Information
Pt		±7.5 dBm
Pr		-28 dBm (Max)
Prx		>-28 dBm
a_{serat} **G.652 D**	1310	0.35 dB/km
	1550	0.28 dB/km
a_{serat} **G.657A**	1310	0.35 dB.km
	1550	0.28 dB.km
Splice Loss (a_s)		0.05 dB/Splice
Connector Loss (a_c)		0.25/Konektor
1:4 Passive Splitter Loss		7.5 dB
Number of Splices		5
Number of Connectors		8
Number of 1:4 Passive Splitters		2
Outdoor Fiber Optic Cable Length		1100 m
Indoor Fiber Optic Cable Length		10 - 60 m
Power Margin		>0 dB

Based on the data above, the link power budget can be calculated as follows:

Downlink Calculation

Total Attenuation Calculation

$a_{tot} = L.a_{serat} + N_c.a_c + N_s.a_s + a_{Sp}$ (1)

a_{tot} = (1,181 km x 0,28 dB/km) + (8 x 0,25) + (5 x 0,05) + (2 x 7,5)

a_{tot} = 17,6 dB

Received Power Calculation:

$P_r = P_t - a_{tot}$ (2)

P_r = 7.5 – 17.6

P_r = -10.1 dBm

Power Margin Calculation

$M = (P_{tx} - P_{rx}) - a_{tot} - M_s$ m (3)

M = (7.5 – 28) – 17.6 – 3

M = 5.8

Uplink Calculation

Total Attenuation Calculation

$a_{tot} = L.a_{serat} + N_c.a_c + N_s.a_s + a_{Sp}$

a_{tot} = (1,181 km x 0,35 dB/km) + (8 x 0,25) + (5 x 0,05) + (2 x 7,5)

a_{tot} = 17,6 dB

Received Power Calculation

$P_r = P_t - a_{tot} - M_s$

P_r = 7.5 – 17.6

P_r = -10.1 dBm

Power Margin Calculation

$M = (P_{tx} - P_{rx}) - a_{tot} - M_s$

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$M = (7.5 - 28) - 17.6 - 3$
$M = 5.8$

Results of Designing and Manufacturing Trainer Modules

After designing and manufacturing the fiber to the home network trainer module, the manufacturing results were obtained starting from the OLT device to the ONT as shown in Figure 5. below.

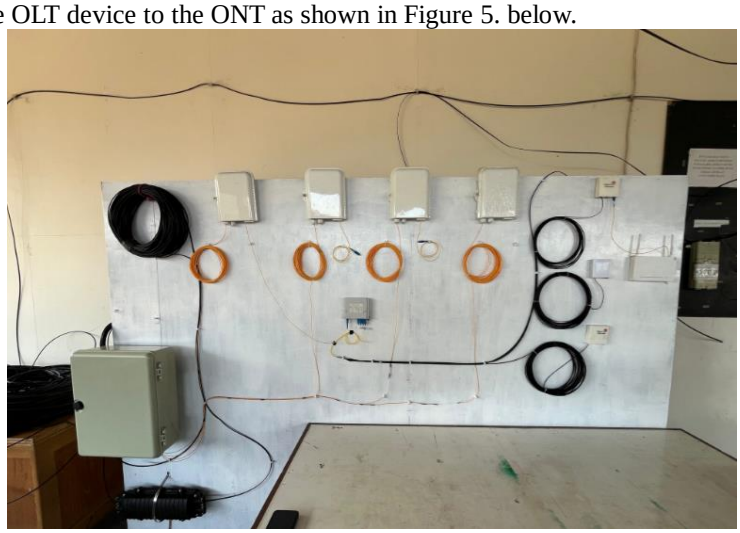


Figure 5. Results of the Trainer Module design

The initial stage of designing and developing this fiber optic–based trainer module begins with the installation of the Optical Distribution Cabinet (ODC) as the starting point for optical signal distribution. It is then connected to a closure, which functions as the joint point between the feeder cable and the distribution cable. From the closure, the signal is transmitted to several Optical Distribution Points (ODPs), which serve as the final distribution points before reaching the end users. This module is also equipped with an external 1:4 splitter, which functions to divide a single optical signal into four outputs, each of which can be delivered to different users through drop core cables. Each drop line is terminated at an optical rosette, which serves as the cable termination point inside the user's premises, and is subsequently connected to the Optical Network Terminal (ONT), a device that receives optical signals and converts them into data signals for internet usage.

Results of Link Power Budget Measurements at the ODC Before the 1:4 Splitter.

At this stage, measurements were conducted at the Optical Distribution Cabinet (ODC) before the signal entered the 1:4 splitter to determine the output power generated by the trainer module. The Optical Line Terminal (OLT) served as the input source, while an Optical Power Meter (OPM) was used to measure the output power produced by the trainer module. The obtained measurement results are presented below.

Output Power Measurement at the OLT

Table 2. Power measurement at the OLT output

Ptx OLT (dBm)	1310 (nm)	1550 (nm)	Ptx OLT (dBm)	1310 (nm)	1550 (nm)
Measurement 1	7.52	7.72	Measurement 6	7.40	7.60
Measurement 2	7.46	7.66	Measurement 7	7.41	7.61
Measurement 3	7.35	7.54	Measurement 8	7.40	7.62
Measurement 4	7.41	7.62	Measurement 9	7.36	7.56
Measurement 5	7.39	7.59	Measurement 10	7.35	7.55

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Based on the output power measurements of the Optical Line Terminal (OLT), the device demonstrated stable and consistent performance at both operating wavelengths. At a wavelength of 1310 nm, the output power ranged from 7.35 dBm to 7.52 dBm, resulting in a variation of 0.17 dBm. Meanwhile, at a wavelength of 1550 nm, the output power ranged from 7.54 dBm to 7.72 dBm, with a variation of 0.18 dBm.

The higher output power observed at 1550 nm compared to 1310 nm is a typical characteristic of optical sources employed in Fiber-to-the-Home (FTTH) systems, where longer wavelengths often produce slightly higher optical power. The average difference between the two wavelengths was approximately 0.20 dBm.

Output Measurement at the ODC Before the 1:4 Splitter (Orange Core/Normal Condition)

Table 3. Output measurement on ODC before splitter (orange core/normal)

Core Orange	**Ptx (dBm) 1310 (nm)**	**Ptx (dBm) 1550 (nm)**	***Core Orange***	**Ptx (dBm) 1310 (nm)**	**Ptx (dBm) 1550 (nm)**
Measurement 1	6.59	6.89	Measurement 6	6.60	6.81
Measurement 2	6.60	6.82	Measurement 7	6.60	6.79
Measurement 3	6.58	6.80	Measurement 8	6.64	6.85
Measurement 4	6.62	6.82	Measurement 9	6.62	6.84
Measurement 5	6.59	6.78	Measurement 10	6.57	6.78

The measurement results presented above represent the evaluation of transmitted optical power (Ptx) in the Optical Distribution Cabinet (ODC) using the orange fiber core. Measurements were carried out at the two standard wavelengths commonly utilized in optical fiber communication systems, namely 1310 nm and 1550 nm.

The results indicate that at 1310 nm, the Ptx values ranged from 6.57 dBm to 6.64 dBm, while at 1550 nm, the values ranged from 6.78 dBm to 6.89 dBm. The variations were relatively small, amounting to only 0.07 dBm and 0.11 dBm, respectively, which indicates excellent transmission power stability.

The stability observed at both wavelengths demonstrates that the optical cable system was operating properly with minimal attenuation. No indications of transmission disturbances, such as weak connections, contaminated connectors, or cable damage, were detected. Therefore, it can be concluded that the ODC orange core was in excellent condition and highly suitable for optical communication applications.

Output Measurement on ODC Before Splitter 1:4 (Green Core/Bending)

Table. 4. Output measurement on ODC before splitter (green core/bending)

Core Orange	**Ptx (dBm) 1310 (nm)**	**Ptx (dBm) 1550 (nm)**	***Core Orange***	**Ptx (dBm) 1310 (nm)**	**Ptx (dBm) 1550 (nm)**
Measurement 1	5.47	5.67	Measurement 6	5.52	5.72
Measurement 2	5.54	5.74	Measurement 7	5.51	5.71
Measurement 3	5.58	5.78	Measurement 8	5.51	5.71
Measurement 4	5.55	5.75	Measurement 9	5.53	5.70
Measurement 5	5.53	5.72	Measurement 10	5.53	5.73

Table 4 presents the results of optical transmit power (Ptx) attenuation measurements at the Optical Distribution Cabinet (ODC) using the green core, which in this context is subjected to bending conditions. The measurements were conducted at two standard wavelengths, namely 1310 nm and 1550 nm.

Based on the table, it can be observed that the transmit power at a wavelength of 1310 nm ranges from 5.47 dBm to 5.58 dBm, while at 1550 nm it ranges from 5.67 dBm to 5.78 dBm. These results indicate that the transmitted signal experiences attenuation; however, it remains within a relatively low range. This attenuation is most likely caused by bending in the optical fiber, which leads to partial light leakage from the fiber core, thereby reducing the measured output power at the cable end.

Bending-induced attenuation tends to have a greater impact at the 1550 nm wavelength due to its longer wavelength and higher sensitivity to fiber curvature. However, in this measurement, the difference in attenuation between 1310 nm and 1550 nm is not significant, indicating that the bending condition remains within the acceptable tolerance limits of the optical network system.

Output Measurement on ODC Before 1:4 Splitter (Blue Core/Broken or Damaged)

Table 5. Output measurements on the ODC before the splitter (blue core/damaged)

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Core Orange	**Ptx (dBm) 1310 (nm)**	**Ptx (dBm) 1550 (nm)**	*Core Orange*	**Ptx (dBm) 1310 (nm)**	**Ptx (dBm) 1550 (nm)**
Measurement 1	2.55	2.73	Measurement 6	2.48	2.69
Measurement 2	2.50	2.78	Measurement 7	2.53	2.72
Measurement 3	2.51	2.73	Measurement 8	2.53	2.73
Measurement 4	2.53	2.75	Measurement 9	2.52	2.72
Measurement 5	2.48	2.71	Measurement 10	2.48	2.71

Table 5 presents the results of optical transmit power (Ptx) attenuation measurements at the Optical Distribution Cabinet (ODC) using the blue core, which in this case is in a broken or damaged condition. The measurements were carried out at two standard wavelengths, namely 1310 nm and 1550 nm.

Based on the data, the transmit power at a wavelength of 1310 nm ranges from 2.48 dBm to 2.55 dBm, while at 1550 nm it ranges from 2.69 dBm to 2.78 dBm. These values are generally lower compared to similar measurements obtained from cores under normal conditions or those experiencing only bending (such as the green core). This indicates that the blue core experiences higher attenuation (loss), most likely due to fiber breakage, cracks, or imperfect splicing.

The high attenuation observed indicates that light transmission within the fiber is significantly disrupted, either due to substantial signal loss at the point of damage or due to abnormal refraction and reflection along the fiber path. This condition can lead to degraded network performance, such as reduced internet speed, intermittent connectivity, or even complete failure of the optical transmission link.

DISCUSSION

In this study, the experimental results obtained from attenuation measurements were compared with simulation results using OptiSystem modeling. The comparison shows a high level of consistency, with a difference of approximately 0.1 dB, indicating that the developed model accurately represents real FTTH GPON network conditions.

Under normal conditions (orange core), the measured transmit power values were stable with minimal variation, confirming that the system operates within standard performance limits and experiences low attenuation. In contrast, under bending conditions (green core), a noticeable decrease in transmit power was observed, demonstrating that physical deformation of the fiber contributes to additional attenuation, although still within acceptable tolerance levels.

Furthermore, under damaged or broken conditions (blue core), the results showed a significant decrease in transmit power and high attenuation levels. This finding highlights that mechanical damage has a major impact on signal degradation, leading to unstable transmission performance. Compared to simulation and theoretical expectations, these results validate that attenuation can be effectively used as an indicator for detecting physical disturbances in optical fiber networks.

Overall, this study successfully demonstrates that the integration of empirical measurement and simulation modeling provides a comprehensive approach to analyzing FTTH network performance. The proposed method not only solves the problem of detecting physical disturbances but also offers a practical contribution as a reliable diagnostic and learning tool in optical communication systems.

CONCLUSION

The design and simulation of the FTTH network in the Optical Fiber Laboratory were successfully carried out through the integration of OLT and ODC devices into a comprehensive trainer module. Simulation using OptiSystem software, combined with real implementation, has created a learning environment that accurately represents the FTTH architecture, in accordance with the practical requirements of Telecommunications Engineering students. The measurement of the link power budget at the output of the FTTH network devices has been successfully conducted and analyzed. The results indicate that the total attenuation under all tested conditions (normal core, bending, and damaged core) remains below the maximum allowable limit of -28 dB. This demonstrates that the designed network system is feasible and meets the required performance standards. The impact of attenuation under various physical conditions in optical fiber has been successfully identified and measured. This study clearly demonstrates that:

a. Under normal conditions (orange core), the attenuation values are stable and consistent.
b. Under bending conditions (green core), there is a significant increase in attenuation, indicating the sensitivity of optical fiber to physical deformation.
c. Under damaged or broken conditions (blue core), the attenuation is the highest and the signal becomes inconsistent, particularly at the 1310 nm wavelength, demonstrating the significant impact of mechanical damage on network performance.

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